AKSHAT NEEMA

Indore, Madhya Pradesh

+91-8435305905 | G-Mail | LinkedIn | GitHub

Summary

Computer Science student at PDPM IIITDMJ specializing in C++, Java, and AI/ML. Proven ability to build high-impact projects, including developing an AI-driven osteoporosis detection system with 93% accuracy and a comprehensive web-based MySQL platform.

Education

PDPM Indian Institute of Information Technology, Design and Manufacturing Jabalpur, MP

B.Tech in Computer Science and Engineering Expected May 2027

- GPA: 8.0/10.0

Skills

- **Languages:** C++, Java, Kotlin, Python, Dart, C
- **Frameworks & Libraries**: Flutter, React, TensorFlow, Keras
- **Web-Technologies:** HTML, CSS, JavaScript, React
- **Databases & Tools**: MySQL, MongoDB, Firestore, Git, GitHub, VS Code, Android Studio, Jupyter Notebook

Projects

AI-Based Osteoporosis Detection System - **Tech:** Python, PyTorch, ResNet-18, Grad-CAM

- Developed a deep learning model using ResNet-18 to detect early signs of osteoporosis from knee X-ray images.
- Applied data preprocessing, augmentation, and regularization techniques to improve model reliability and prevent overfitting.
- Implemented Grad-CAM explainability to highlight key bone regions influencing predictions and improve interpretability.
- Achieved ~93% validation accuracy, demonstrating the potential of AI-driven data analysis for healthcare decision support systems.

Salon Booking Platform - **Tech:** Flutter, Dart, Firebase (Authentication, Firestore)

(GitHub Repository - Private)

- Developed a cross-platform digital platform to streamline appointment scheduling and service management for small service businesses.
- Designed a structured **cloud database using Firestore** to manage customer data, services, and booking transactions with real-time updates.
- Implemented secure **OTP-based authentication and role-based access control** for customers and business operators.
- Built an end-to-end digital workflow that reduced manual scheduling and improved operational efficiency.

Demo Video: Youtube

Achievements & Leadership

- Participated in Adobe Hackathon, gained hands-on exposure to real-world problem solving and team collaboration.
- Working on the Hostel Management module of the institute's Fusion ERP system, implementing backend logic and database management for hostel operations.
- Attended NxtWave – AI Workshop, learning practical applications of Artificial Intelligence.
- Active participant in coding competitions and problem-solving platforms, improving algorithmic thinking.

Certifications

- HP Power Lab
- Adobe India Hackathon
- Personal Branding - Pod